
Data Science Practical Viva Question Bank (1-10)

Practical 1: Data Wrangling I

Concepts: Loading, Cleaning, and Initial Transformation.

1. **What is Pandas?** A library for data manipulation and analysis using DataFrames.
2. **What is a DataFrame?** A two-dimensional, size-mutable, tabular data structure with labeled axes.
3. **Purpose of `read_csv()`?** To load a CSV file into a Pandas DataFrame.
4. **What does `isnull().sum()` do?** It identifies missing values and provides a count for each column.
5. **What is the use of `describe()`?** It provides a statistical summary (mean, std, min, max, etc.) of numeric columns.
6. **Difference between `head()` and `tail()`?** `head()` shows the first 5 rows; `tail()` shows the last 5.
7. **What are Categorical variables?** Variables that represent discrete types or groups (e.g., Color, Gender).
8. **What is Data Normalization?** Scaling numerical features to a standard range, usually 0 to 1.
9. **How do you change a column's data type?** By using the `.astype()` function.
10. **What is One-Hot Encoding?** Converting categorical text into binary columns (0 and 1).
11. **Explain the "Dummy Variable Trap."** A scenario where independent variables are highly correlated (multi-collinearity).
12. **Why use `drop_first=True` in `get_dummies`?** To remove one redundant column and prevent the Dummy Variable Trap.
13. **What is `df.shape`?** It returns the total number of rows and columns as a tuple.
14. **Difference between `loc` and `iloc`?** `loc` uses labels to access data; `iloc` uses integer indexes.
15. **What is a Series?** A one-dimensional labeled array capable of holding any data type.
16. **Why is data wrangling necessary?** Raw data is often incomplete or inconsistent; wrangling makes it ready for modeling.
17. **What is Data Formatting?** Ensuring data is in a consistent structure (e.g., converting "05-2026" to a Date object).
18. **How to remove duplicate rows?** Using the `df.drop_duplicates()` method.
19. **What is NumPy?** A library for numerical computing and performing operations on large arrays.
20. **What are logical variables?** Boolean values (True or False) used for filtering or conditions.

Practical 2: Data Wrangling II (Outliers & Missing Values)

Concepts: Imputation, IQR Method, and Skewness.

1. **What are outliers?** Data points that differ significantly from the rest of the observations.
 2. **What is the IQR method?** A technique to find outliers using the Interquartile Range ($Q3 - Q1$).
 3. **What is the formula for Outlier detection?** Lower Bound: $Q1 - 1.5(IQR)$; Upper Bound: $Q3 + 1.5(IQR)$.
 4. **What is Data Imputation?** The process of replacing missing data with substituted values (mean/median/mode).
 5. **When is Median better than Mean for imputation?** When the data contains strong outliers that skew the average.
 6. **What is Skewness?** A measure of the asymmetry of the probability distribution of a variable.
 7. **How does Log Transformation help?** It compresses high-value ranges and expands low-value ranges to reduce skewness.
 8. **What is `np.where()`?** A function used to apply a condition; if True, do X; if False, do Y.
 9. **What is Data Consistency?** Ensuring data follows logical rules (e.g., a "Price" column cannot be negative).
 10. **What is Min-Max Scaling?** A normalization technique: $(x - \min) / (\max - \min)$.
 11. **What is a Z-Score?** It tells you how many standard deviations a data point is from the mean.
 12. **What does a Boxplot represent?** It visually shows the median, quartiles, and outliers of a dataset.
 13. **Why use `fillna()`?** To replace all NaN (missing) values with a specific value or strategy.
 14. **What is "Capping"?** Replacing outliers with the maximum or minimum "normal" values.
 15. **What is a Normal Distribution?** A symmetric bell curve where mean, median, and mode are equal.
 16. **Difference between Univariate and Bivariate outliers?** Univariate is an outlier in one variable; Bivariate is an outlier across two.
 17. **How to check skewness in Pandas?** Using the `df.skew()` method.
 18. **Why are outliers a problem?** They can lead to inaccurate models and biased statistical results.
 19. **What is Data Transformation?** Applying a mathematical function to data to change its distribution.
 20. **What is StandardScaler?** A tool to scale data to have a mean of 0 and a variance of 1.
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Practical 3: Descriptive Statistics

Concepts: Central Tendency and Measures of Variability.

1. **What is the Mean?** The sum of all values divided by the total count.
2. **What is the Median?** The middle value when data is sorted in ascending order.
3. **What is the Mode?** The value that appears most frequently in a dataset.
4. **Define Standard Deviation.** It measures the spread of data points around the mean.
5. **What is Variance?** The average of the squared differences from the Mean.
6. **How does `groupby()` work?** It splits data into groups based on a key for independent analysis.
7. **What is `.agg()`?** A function used to apply multiple statistical operations at once.
8. **What is a Percentile?** A score below which a given percentage of observations fall.
9. **What is Qualitative data?** Non-numeric information (e.g., name, color, species).
10. **What is Quantitative data?** Numeric information that can be measured (e.g., height, score).
11. **What does a low Standard Deviation mean?** Data points are close to the mean (consistent).
12. **Difference between Population and Sample?** Population is the whole group; Sample is a small part of it.
13. **What is the Range?** The difference between the maximum and minimum values.
14. **How to describe statistics for one category?** Use `df.groupby('category').describe()`.
15. **What is the 75th Percentile (`Q3`)?** The value below which 75% of the data lies.
16. **What is `df.nunique()`?** It counts the number of unique entries in each column.
17. **What is a Frequency Table?** A table showing how many times each value occurs.
18. **What is Kurtosis?** A measure of whether the data is heavy-tailed or light-tailed compared to a normal distribution.
19. **How to find correlation?** Use the `df.corr()` method.
20. **What is Nominal data?** Categorical data with no logical order (e.g., City names).

Practical 4: Linear Regression

Concepts: Predicting continuous values and Model Evaluation.

1. **What is Linear Regression?** A supervised learning algorithm used to predict a continuous target variable.
2. **What is the equation of a line?** $y = mx + c$ (where m is slope and c is intercept).
3. **What is the Target variable?** The variable you are trying to predict (the y variable).
4. **What are Features?** The independent variables used to make predictions (the X variables).
5. **What is Train-Test Split?** Reserving a portion of data to test how well the model performs on unseen data.
6. **Explain Mean Squared Error (MSE).** The average of the squares of the differences between actual and predicted values.

7. **What is the R^2 Score?** It indicates how much of the variance in y is explained by X (0 to 1).
8. **What is Overfitting?** When a model performs perfectly on training data but poorly on test data.
9. **What is Underfitting?** When a model is too simple to learn the patterns in the data.
10. **What is the Intercept?** The value of y when X is zero.
11. **What is the Coefficient?** It represents the weight or impact of a feature on the target variable.
12. **What is a Residual?** The difference between an actual value and its predicted value ($\text{Actual} - \text{Predicted}$).
13. **Why use `random_state`?** To ensure the same data split is produced every time the code is executed.
14. **Is Linear Regression used for labels?** No, it is strictly for numerical/continuous values.
15. **What is Multiple Linear Regression?** Linear regression using two or more independent features.
16. **What is Multi-collinearity?** When two independent variables are very highly correlated with each other.
17. **How do you evaluate a regression model?** Using metrics like MSE, MAE, and R^2 .
18. **What is the "Line of Best Fit"?** The line that minimizes the sum of squared residuals.
19. **What is Scikit-learn?** The primary Python library for Machine Learning algorithms.
20. **What does `model.fit(X, y)` do?** It trains the model to find the optimal coefficients for the data.

Practical 5: Logistic Regression

Concepts: Binary Classification and the Confusion Matrix.

1. **What is Logistic Regression?** A classification algorithm used to predict binary outcomes (0 or 1).
2. **What is the Sigmoid Function?** A function that maps any real-valued number into a range between 0 and 1.
3. **What is a Confusion Matrix?** A table showing the performance of a classification model.
4. **What is a True Positive (TP)?** The model correctly predicted the positive class.
5. **What is a False Positive (FP)?** The model predicted positive, but it was actually negative.
6. **What is a True Negative (TN)?** The model correctly predicted the negative class.
7. **What is a False Negative (FN)?** The model predicted negative, but it was actually positive.
8. **Define Accuracy.** The ratio of correct predictions to the total number of predictions.
9. **Define Precision.** The accuracy of positive predictions ($\text{TP} / (\text{TP} + \text{FP})$).
10. **Define Recall (Sensitivity).** The ability of the model to find all positive cases ($\text{TP} / (\text{TP} + \text{FN})$).

11. **What is the Error Rate?** The percentage of incorrect predictions ($1 - \text{Accuracy}$).
12. **When is Precision critical?** When the cost of a False Positive is high (e.g., marking a safe email as spam).
13. **When is Recall critical?** When the cost of a False Negative is high (e.g., missing a disease diagnosis).
14. **What is the Decision Boundary?** The threshold (usually 0.5) that determines which class a probability belongs to.
15. **What is the F1-Score?** The weighted average (harmonic mean) of Precision and Recall.
16. **Is Logistic Regression linear?** It uses a linear combination of inputs but applies a non-linear sigmoid function.
17. **What are "Log-Odds"?** The natural logarithm of the odds of an event occurring.
18. **Why is scaling important for Logistic Regression?** To help the optimization algorithm (Gradient Descent) converge faster.
19. **What is Binary Classification?** Classification with exactly two possible outcomes.
20. **What is the difference between `predict()` and `predict_proba()`?** `predict` gives the class (0 or 1); `predict_proba` gives the probability (e.g., 0.85).

Practical 6: Naïve Bayes

Concepts: Probability and Feature Independence.

1. **What is Naïve Bayes?** A classification algorithm based on Bayes' Theorem.
2. **Why is it called "Naïve"?** Because it assumes all input features are independent of each other.
3. **What is Bayes' Theorem?** $P(A|B) = [P(B|A) \cdot P(A)] / P(B)$.
4. **What is Gaussian Naïve Bayes?** A variant of Naïve Bayes used when features are continuous and follow a normal distribution.
5. **What is Multinomial Naïve Bayes?** Used for discrete data, common in document classification.
6. **What is Bernoulli Naïve Bayes?** Used for features that are binary (True/False).
7. **What is Prior Probability?** The initial probability of a class before observing the data.
8. **What is Likelihood?** The probability of seeing the features given a specific class.
9. **What is a major advantage of Naïve Bayes?** It is extremely fast and works well with high-dimensional data.
10. **What is a major disadvantage?** The independence assumption is rarely true in real-world data.
11. **What is the "Zero Frequency" problem?** If a category was never seen in training, its probability becomes zero.
12. **What is Laplace Smoothing?** Adding a small value (usually 1) to counts to avoid zero probabilities.
13. **What is Posterior Probability?** The final probability calculated after considering the evidence.
14. **Does Naïve Bayes require a lot of data?** No, it is quite efficient even with smaller datasets.

15. **What is the Iris dataset?** A collection of flower measurements used to predict species.
16. **Name the three species in Iris.** Setosa, Versicolor, and Virginica.
17. **What is "Macro" averaging in metrics?** Calculating the metric for each class separately and then taking the average.
18. **Difference between Naïve Bayes and Logistic Regression?** NB is a generative model; LR is a discriminative model.
19. **Can Naïve Bayes handle text?** Yes, it is one of the most popular algorithms for text classification.
20. **What is the probability of a specific class?** The class with the highest posterior probability is the prediction.

Practical 7: Text Analytics

Concepts: Tokenization, NLP, and TF-IDF.

1. **What is NLP?** Natural Language Processing; the study of how computers interact with human language.
2. **What is Tokenization?** Splitting a string of text into smaller units called "tokens" (usually words).
3. **What are Stop Words?** Common words (e.g., "the", "is", "a") that are removed to focus on meaningful words.
4. **What is Stemming?** A crude process of removing suffixes to reach the root form of a word.
5. **What is Lemmatization?** A process that uses a dictionary to find the linguistically correct root (lemma).
6. **What is the difference between Stemming and Lemmatization?** Lemmatization is slower but smarter; Stemming is fast but can produce non-words.
7. **What is POS Tagging?** Labeling words as Nouns, Verbs, Adjectives, etc.
8. **What is Term Frequency (TF)?** How many times a word appears in a single document.
9. **What is Inverse Document Frequency (IDF)?** It reduces the weight of words that appear very frequently across all documents.
10. **What is TF-IDF?** A numerical statistic that reflects how important a word is to a document in a collection.
11. **What is NLTK?** The Natural Language Toolkit library for Python.
12. **What is a Corpus?** A large and structured set of texts used for statistical analysis.
13. **What is WordNet?** A large lexical database of English used for finding synonyms and lemmas.
14. **What is "Bag of Words"?** A simplified representation where text is seen as an unordered collection of words.
15. **Why do we convert text to lowercase?** To treat "Apple" and "apple" as the same token.
16. **Why remove punctuation?** To prevent punctuation from being treated as a unique word/token.

17. **What is TfidfVectorizer?** A Scikit-learn tool that combines `CountVectorizer` and `TfidfTransformer`.
18. **What is Sentiment Analysis?** Using NLP to determine if a text is positive, negative, or neutral.
19. **What is an N-gram?** A sequence of n words (e.g., "New York" is a bigram).
20. **What is Named Entity Recognition (NER)?** Identifying proper names of people, places, or organizations in text.

Practical 8, 9 & 10: Data Visualization

Concepts: Seaborn, Matplotlib, and Pattern Analysis.

1. **What is Seaborn?** A visualization library built on Matplotlib that provides a high-level interface.
2. **What is Matplotlib?** The core plotting library in Python for creating static, animated, and interactive visualizations.
3. **What is a Histogram?** A graph used to show the frequency distribution of a continuous numeric variable.
4. **What is a Boxplot?** A graph that shows the distribution of data based on a five-number summary.
5. **What is a Scatter Plot?** A plot that uses dots to represent values for two different numeric variables.
6. **What is a Heatmap?** A way to visualize a matrix of numbers using colors.
7. **What is a Countplot?** A bar chart that shows the number of occurrences of each category.
8. **What is the `hue` parameter in Seaborn?** It allows you to color data points based on a categorical variable.
9. **What is a KDE Plot?** A plot that represents the probability density of continuous data.
10. **Difference between Numeric and Nominal variables?** Numeric is numbers; Nominal is categories with no order.
11. **How to save a plot as an image?** Using `plt.savefig('my_plot.png')`.
12. **Why do we use visualizations?** To identify trends, patterns, and outliers that numbers alone cannot show.
13. **What is a Bimodal Distribution?** A distribution that has two distinct peaks.
14. **What are "bins" in a histogram?** The number of ranges/buckets the data is split into.
15. **What is a Pairplot?** A grid of scatter plots showing the relationship between every pair of variables.
16. **What is a Violin Plot?** A combination of a boxplot and a KDE plot.
17. **What are the "whiskers" in a Boxplot?** Lines that extend to show the variability outside the upper and lower quartiles.
18. **How do you set a plot title?** Using `plt.title("My Title")`.
19. **What is a Bar Chart used for?** Comparing values across different categories.
20. **Which feature in Iris usually has outliers?** Sepal Width usually shows visible outliers in a boxplot.

Good luck with your exam tomorrow, Vikas!